ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE I · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice i.

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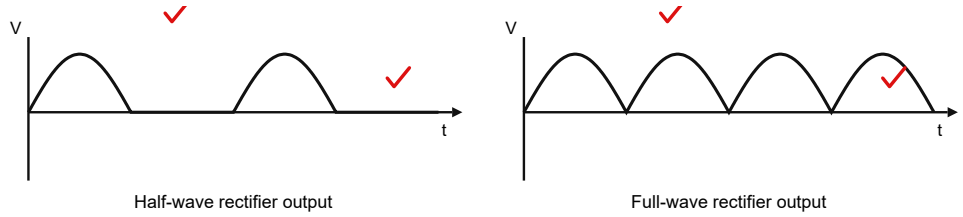
How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Circuits may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 Any TWO types ✓✓: ceramic; electrolytic; tantalum; polyester / film; variable. Any ONE use ✓: smoothing a power supply; timing; filtering noise; coupling signals; motor suppression. (3)

1.2 $\tau = R \times C = 100\,000 \times 47 \times 10^{-6}$ ✓ $= 4{,}7$ s ✓; fully charged after about 5τ ✓ $= 23{,}5$ s ✓ (4)

1.3



Half-wave rectifier output Full-wave rectifier output

(Half-wave: one diode passes only one half of each AC cycle, so the output has gaps 2; waveform – positive half-cycles only with gaps 1. Full-wave: a bridge (or centre-tapped transformer and two diodes) uses both halves 2; waveform – continuous positive humps 1)

(6)

1.4 It has seven LED segments (a–g) arranged in a figure 8 ✓; different combinations of segments are switched on ✓ to form the digits 0–9, controlled by a decoder or microcontroller ✓. (3)

1.5 The component produces heat ✓; the heat sink conducts the heat away and releases it into the air so the component does not overheat and fail ✓. (2)

1.6 The time taken for one complete cycle of the waveform ✓, measured in seconds (T = 1 ÷ f) ✓. (2)

[20]

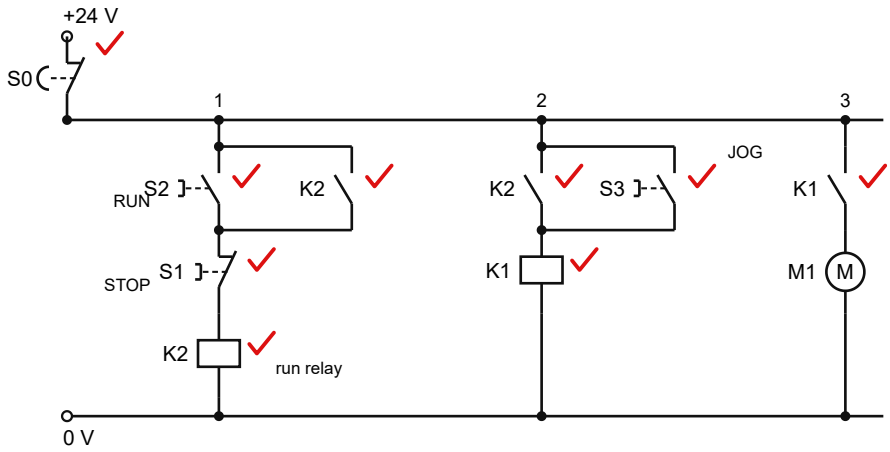
QUESTION 2

2.1 They replace mechanical cams with a programmed position profile ✓ so that a follower axis (motor) moves in step with a master axis according to the profile ✓; the profile can be changed in software for a new product without changing hardware ✓. (3)

2.2 Any THREE ✓✓✓: oscillator / PWM generator (e.g. 555 timer or microcontroller); duty-cycle control (potentiometer); switching transistor / MOSFET (driver); flyback diode; power supply. (3)

2.3 Any THREE ✓✓✓: changing the supply (armature) voltage; PWM; changing the field current (shunt motors); a series resistance in the armature circuit; a gearbox (speed of the load). (3)

2.4



(Supply lines 1; E-stop (NC) 1 and STOP (NC) 1 in series; RUN (NO) 1 latching a run relay K2 1; JOG (NO) in parallel with the K2 contact, driving K1 directly without the latch 1; K1 switching the motor 1; correct symbols 1. Other correct jog / run circuits are accepted.)

(8)

2.5 A system that controls the speed, torque or position of an electric motor ✓, made up of the motor, a power converter / controller and its control electronics ✓, used to drive a machine ✓. (3)

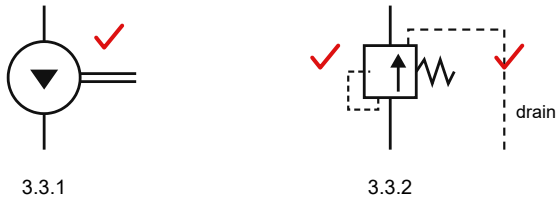
[20]

QUESTION 3

3.1 Any THREE ✓✓✓: gear pump (external / internal); vane pump; piston pump (axial / radial); screw pump. (3)

3.2 Two meshing gears turn inside a close-fitting housing ✓; as the teeth come out of mesh at the inlet, a low pressure draws oil in ✓; the oil is carried around the outside of the gears in the spaces between the teeth ✓; where the teeth mesh again, the oil is forced out of the outlet ✓. (4)

3.3

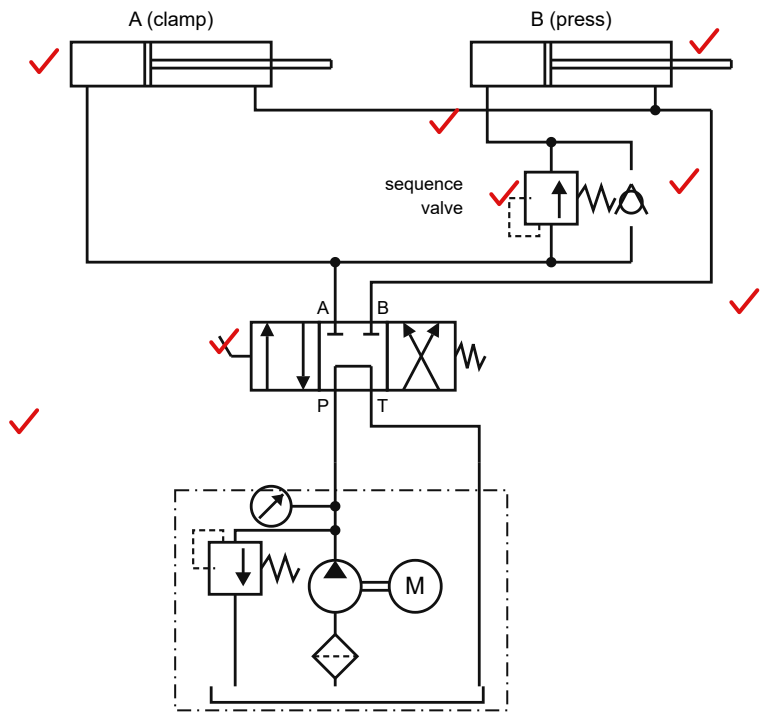


3.3.1 3.3.2

(3.3.1: circle with ONE solid triangle pointing INWARDS 1 and a drive shaft 1. 3.3.2: square valve with an offset arrow and adjustable spring 1, a dashed pilot line from the inlet 1, and an external drain line 1)

(5)

3.4



(Power pack with relief valve 1; 4/3 manual valve 1; cylinder A connected so it extends first 1; sequence valve in the line to cylinder B 1 with its pilot from the clamp (A) line 1; cylinder B connected after the sequence valve 1; return lines to tank with a check valve so B can retract freely 1; correct symbols, neat 1)

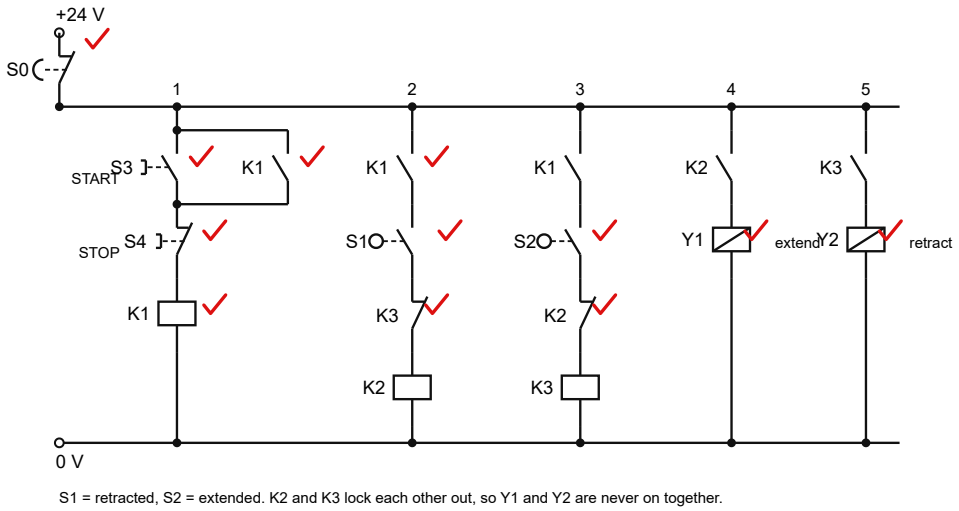
(8)

[20]

QUESTION 4

4.1 A PNP sensor switches the positive supply to its output (sourcing): the output goes to +24 V when it detects ✓✓. An NPN sensor switches its output to 0 V (sinking): the load is connected between + and the output ✓✓. (4)

4.2



S1 = retracted, S2 = extended. K2 and K3 lock each other out, so Y1 and Y2 are never on together.

(Supply lines 1; E-stop in the supply 1; START (NO) 1 with relay K1 latching 1 through STOP (NC) 1 (K1 = "auto running"); K1 contact and S1 (retracted) switching Y1 (extend) 2; K1 contact and S2 (extended) switching Y2 (retract) 2; Y1 and Y2 interlocked so they cannot be on together 1; on STOP the cylinder finishes its stroke safely 1; correct symbols and neat 1)

(12)

4.3 Any FOUR ✓✓✓✓: very large forces from compact cylinders; precise control with sensors and PLCs; can be automated and sequenced easily; remote operation; smooth, controllable speed; safety interlocks are easy to add; overload protection with relief valves. (4)

[20]

QUESTION 5

5.1 Any THREE ✓✓✓: worn or leaking piston seals (internal leakage); leaking directional valve spool (oil passes back to tank); faulty or missing pilot-operated check / counterbalance valve; external leak at fittings; relief valve leaking. (3)

5.2 Any FIVE ✓✓✓✓✓: cracks or cuts in the outer cover; bulges or blisters; leaks at the fittings; kinks or twists; chafing / rubbing against parts; correct pressure rating and length; loose or damaged fittings; signs of heat damage; date of manufacture / age. (5)

5.3 With the system at low pressure, operate the cylinder slowly through several full strokes ✓; open the bleed screws at the high points (or loosen a fitting slightly) until oil without bubbles comes out ✓; tighten the bleed points ✓; top up the tank and check that the actuators move smoothly without sponginess or noise ✓. (4)

5.4

FAULT	POSSIBLE CAUSE	REMEDY
Pump is noisy	Air in the oil / low oil level	Top up the oil and bleed the air ✓✓
Cylinder moves slowly	Flow control set too low / worn pump	Adjust the flow control / repair or replace the pump ✓✓

(Correct column headings 2; two sensible rows 4. Other sensible faults are accepted.)

(6)

5.5 Different oils have different viscosities and additives ✓; the wrong oil can damage seals, cause overheating and poor performance ✓. (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE